

Geographic sampling and composition of the restaurant-review dataset

Abstract

Recommendation quality is constrained by the catalogue and interaction data available to the model. We assembled a US restaurant dataset by searching Census block groups (CBGs) selected for high foot traffic, high population density, and surrounding geographic coverage. Restaurant metadata came from Google Places API (New); public review pages were collected separately and processed into a canonical analytical table. The resulting dataset contains 75,203 restaurants and 5,544,947 reviews. This article documents the sampling frame, collection safeguards, cuisine normalization, demographic diagnostics, and the chronological evaluation population used by the subsequent recommendation experiments.

Geographic sampling and restaurant discovery

A CBG is a small Census statistical geography. We used CBGs because they provide a more precise unit of coverage than states or cities and can be linked to population and spatial-neighborhood information. The discovery frame combined three signals:

Coverage component	Purpose
High-foot-traffic CBGs	Represent active commercial and dining areas
Top 10,000 population-density CBGs	Avoid omitting dense residential and mixed-use areas
Surrounding CBGs and H3 search cells	Cover restaurants near CBG boundaries and local neighborhoods

Every selected polygon was covered with H3 resolution-9 search cells. Google Places API (New) Nearby Search supplied place identity and metadata, and Google Place ID served as the restaurant deduplication key. Terminal markers were written for successful searches and confirmed empty searches. This distinction prevents a large CBG with no returned restaurants from being searched again merely because its result set is empty.

The static map below shows 11,992 terminally searched CBG identifiers that map directly to 2024 boundaries. Another 1,343 terminal identifiers come from an earlier Census boundary vintage and are omitted from the static geometry rather than assigned to an incorrect polygon. The map is a search-footprint visualization, not a restaurant-density map.

Geographic coverage of US restaurant discovery

11,992 terminally searched CBGs mapped to 2024 boundaries; coverage combines population density, foot traffic, and surrounding areas

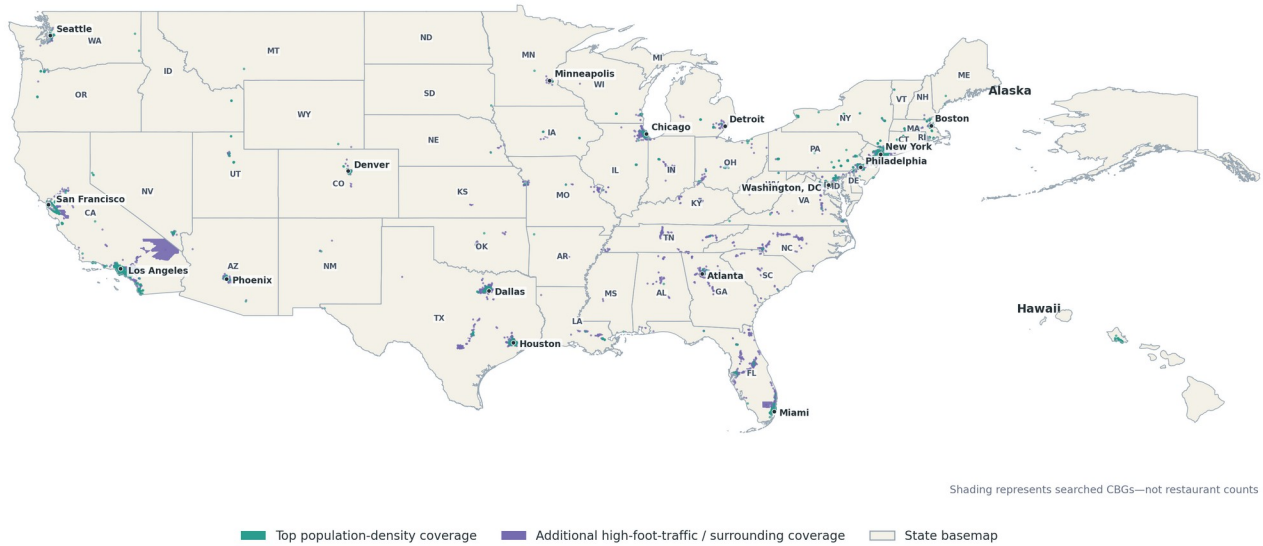


Figure 1. Terminally searched Census block groups. Teal marks top population-density coverage; purple marks additional high-foot-traffic and surrounding coverage. Alaska and Hawaii are shown as insets.

Final analytical corpus

The restaurant catalogue spans 49 state/DC jurisdictions; Alaska and Montana have no retained restaurant record. Coverage is therefore broad but not a probability sample of US restaurants. Google visibility, the selected CBGs, and the query geometry all influence inclusion.

Dataset quantity	Count
Unique restaurants	75,203
Raw review objects	5,545,260
Canonical analytical reviews	5,544,947
Identified reviewers	3,930,034
Restaurants with at least one analytical review	70,625
Reviewers eligible for modeling (4+ restaurants)	156,261

Restaurant catalogue coverage and supported cuisine composition

75,203 restaurants; cuisine panel excludes 6,996 venues with insufficient evidence

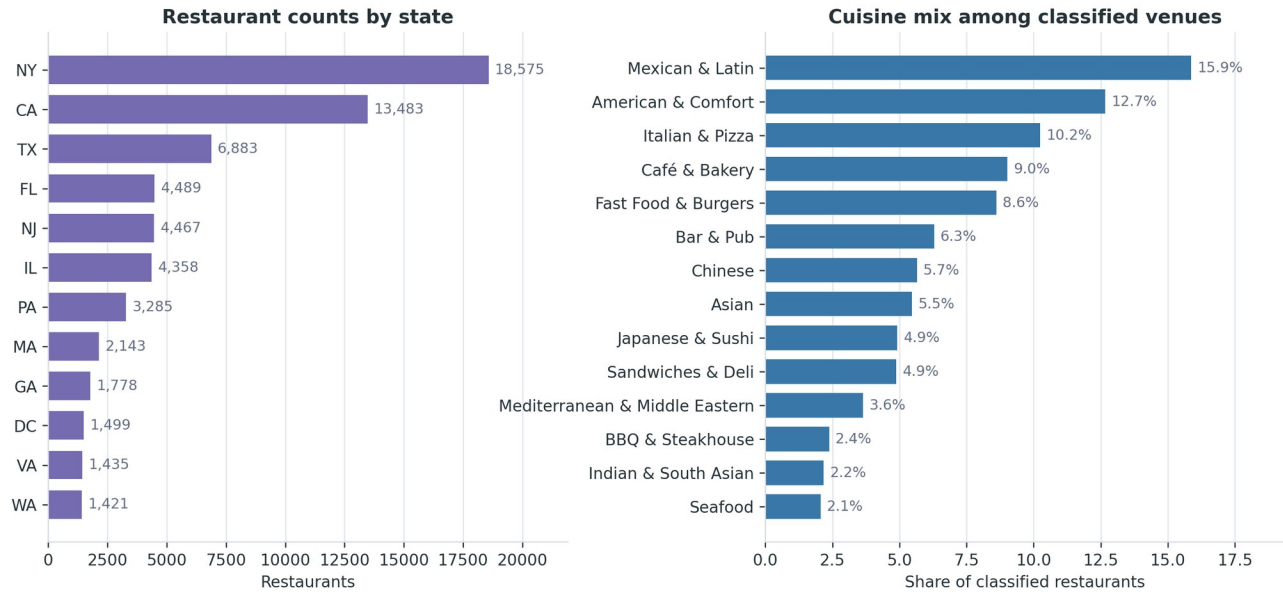


Figure 2. State coverage and cuisine distribution. Cuisine shares exclude venues without a supported cuisine label, but those venues remain in the analytical and modeling tables.

Review collection used Camoufox with reusable browser contexts and a fresh page per restaurant. Starts were staggered and rate-limited, and scrolling was paced to allow dynamically rendered review panels to load. A known-positive probe distinguished a genuine absence of reviews from connectivity failures, delayed rendering, and automated-traffic challenges. Retryable failures were retained as retryable states rather than converted into false “no review” outcomes.

Review-rating and reviewer-activity distributions

The corpus is strongly positive: 63.2% of reviews have five stars and 12.5% have four stars. Reviewer activity is highly skewed in the opposite direction: 82.0% of identified reviewers occur at only one restaurant. The model population therefore represents a denser behavioral subset, not a miniature copy of all reviewers.

Review-rating distribution and modeling-population threshold

The leave-two-out experiment retains 156,261 users with at least four unique restaurants

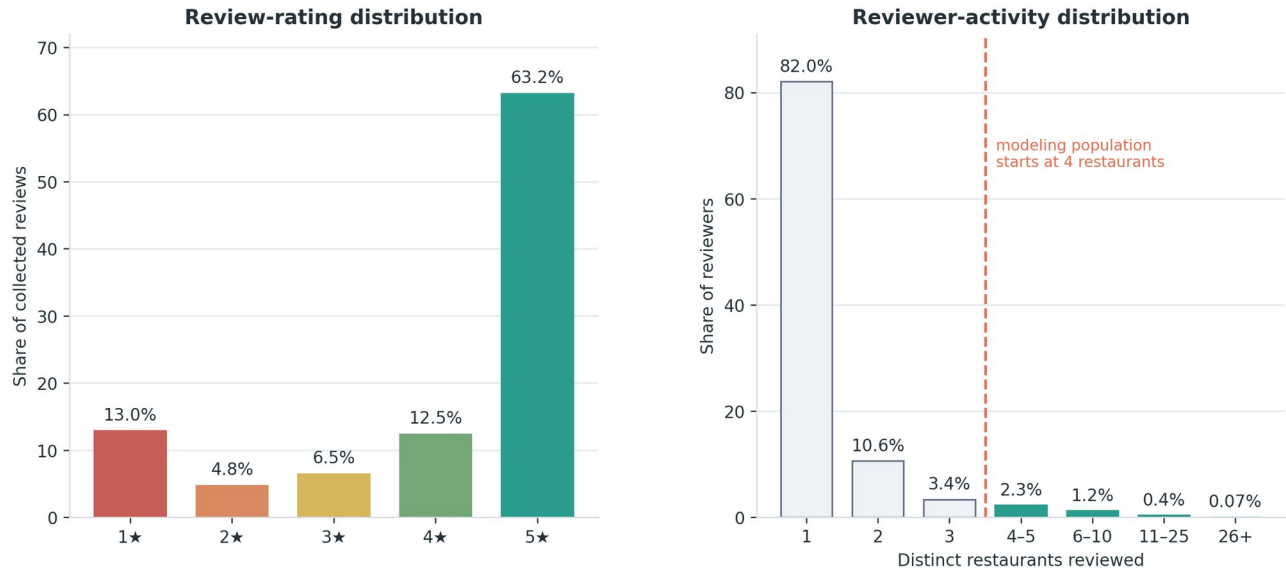


Figure 3. Rating distribution and reviewer-activity funnel. The modeling threshold begins at four distinct restaurants because each user must retain training history after validation and test interactions are held out.

Distributional characteristics of 156,261 modeling users

19.6% of reviews · 99% Local Guides · 2.0 states each on average · 36% sample 5+ cuisines

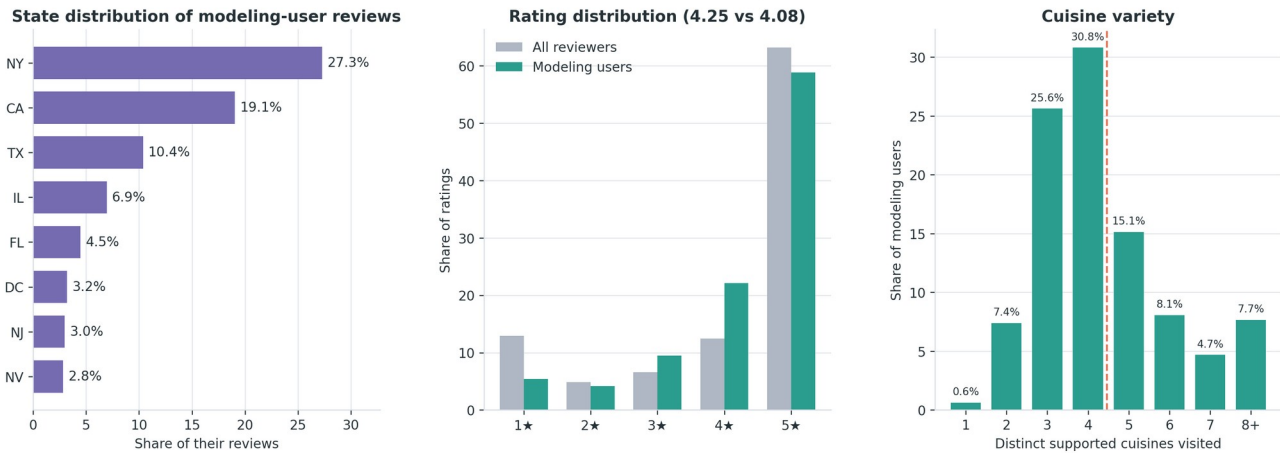


Figure 4. Geography, rating behavior, and cuisine breadth among the 156,261 modeling users.

Review content varies with sentiment. Critical reviews contain more text, while attached photos are more common in positive reviews. These differences matter for semantic feature extraction: negative reviews provide detailed failure descriptions, whereas positive reviews more often add visual evidence.

Review text and photo availability by star rating

Long complaints and photo-heavy praise give the LLM different kinds of evidence

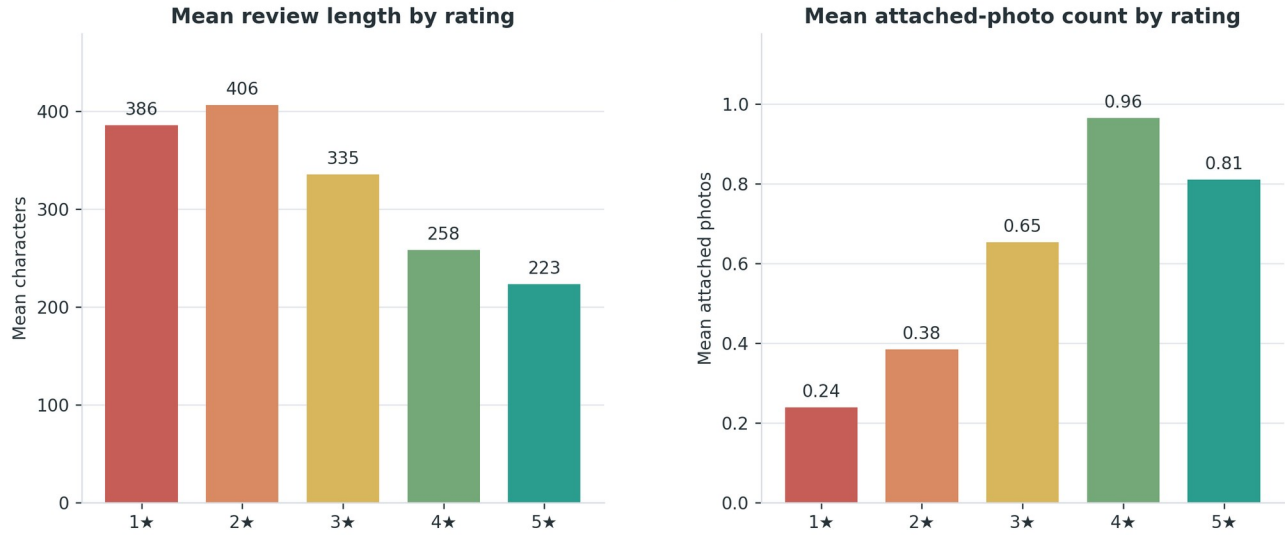


Figure 5. Mean review length and attached-photo count by star rating.

Cuisine normalization

Google's broad type mapping assigned 15,747 restaurants to a generic category. A targeted local 27B model re-examined only those records using names, place types, summaries, and available review evidence. It assigned 8,751 supported specific labels; 6,996 restaurants retained an insufficient-evidence state. Cuisine figures omit the latter group rather than forcing a label. The restaurants themselves remain in the dataset and model candidate universe.

State distribution and cuisine composition of the modeling catalogue

Stacked bars use the ten most common supported cuisines; insufficient-evidence venues are omitted

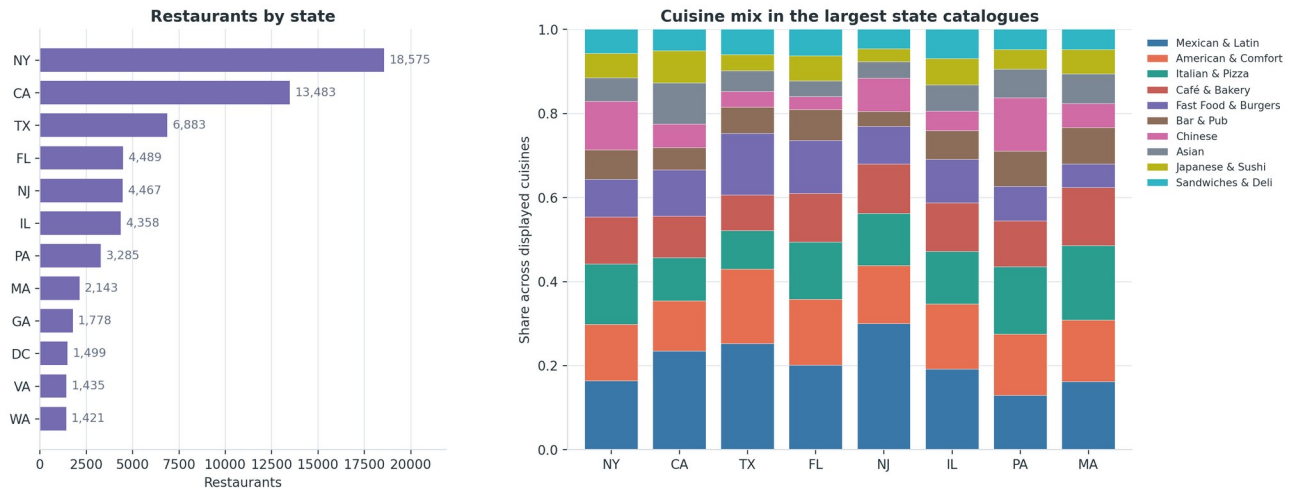


Figure 6. Restaurant counts and cuisine composition in the largest state catalogues. The stacked panel uses supported cuisine labels only.

Demographic rating analysis

Gender and race/ethnicity labels are probabilistic name-inference outputs, not self-reported identity. They are used only for descriptive diagnostics and subgroup evaluation. They should not be interpreted as population estimates or used for sensitive targeting.

Across classified review rows, female-coded reviewers average 4.1359 stars and male-coded reviewers average 4.0974; both medians are 5 stars. Repeated reviews from one person violate the independence assumption of an ordinary review-level test, so the primary significance analysis uses one mean rating per reviewer. At that level, the means are 4.0969 and 4.0461, respectively, and both medians remain 5.

Analysis level	Female mean	Male mean	Female median	Male median	Difference
Review rows	4.1359	4.0974	5.0	5.0	+0.0385
One mean per reviewer	4.0969	4.0461	5.0	5.0	+0.0508

The reviewer-level Welch test gives $t=30.09$, $p=8.0 \times 10^{-199}$, with a 95% confidence interval of [0.0475, 0.0541] stars for the mean difference. The effect is statistically detectable because the sample contains nearly 2.92 million classified reviewers, but it is substantively negligible: Cohen's $d=0.035$. Statistical significance should not be confused with practical importance.

Mean rating by inferred gender and restaurant cuisine

Overall difference is statistically detectable but negligible (reviewer-level Welch $p < 10^{-198}$; $d=0.035$); labels are name-inferred

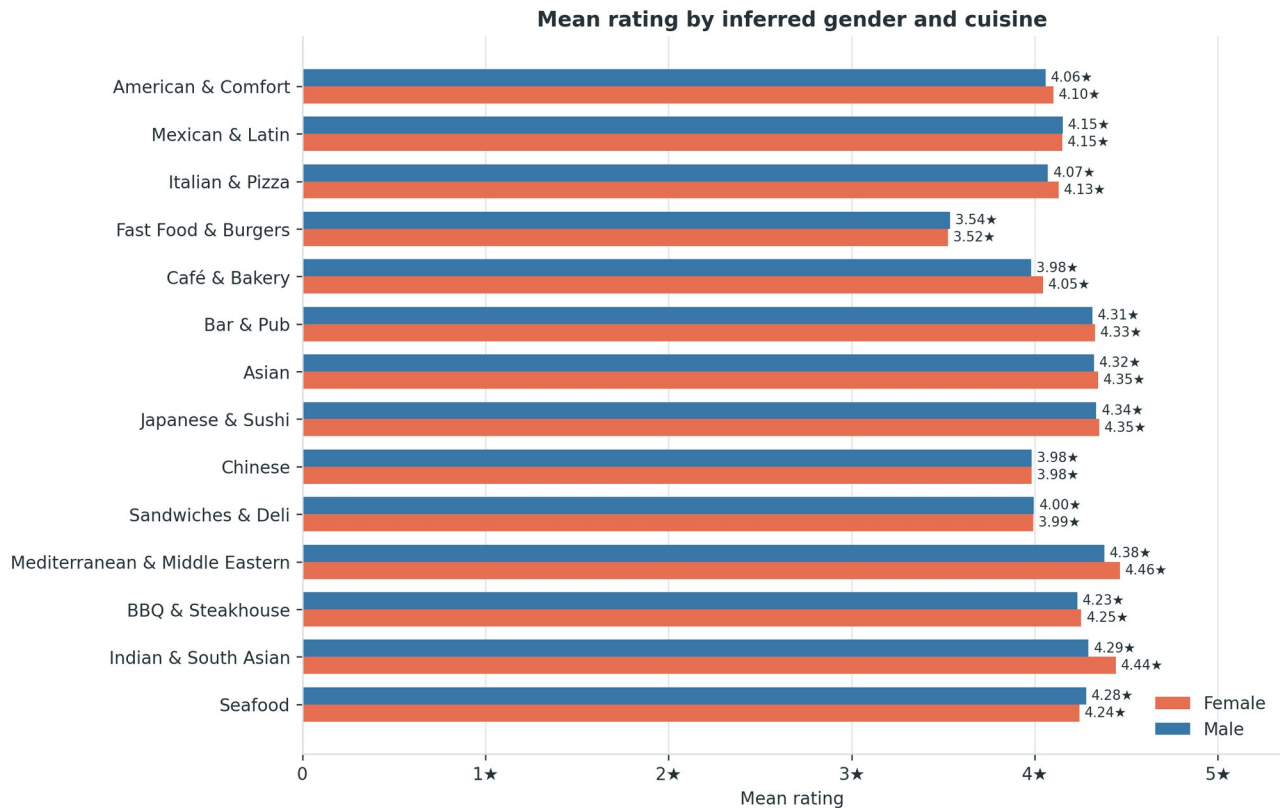


Figure 7. Mean rating by cuisine and inferred gender. Bars use the full zero-to-five-star scale.

Mean ratings by inferred race/ethnicity and cuisine

Descriptive name-inference diagnostic; AIAN omitted because the sample is very small

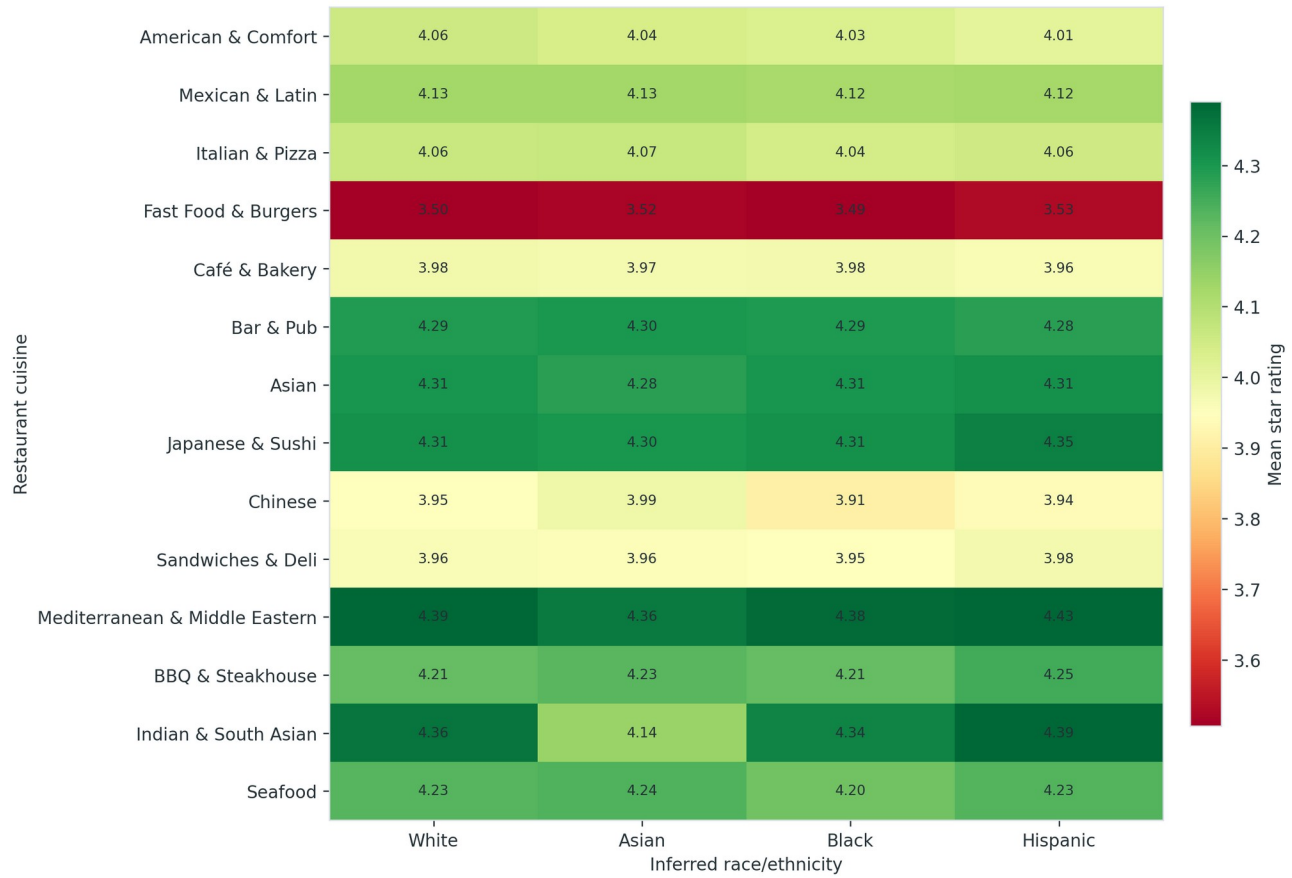


Figure 8. Cuisine-conditioned mean ratings for sufficiently represented inferred race/ethnicity groups.

Review composition by inferred demographic group and cuisine

Shares are calculated within classified demographic subsets and should not be read as population estimates

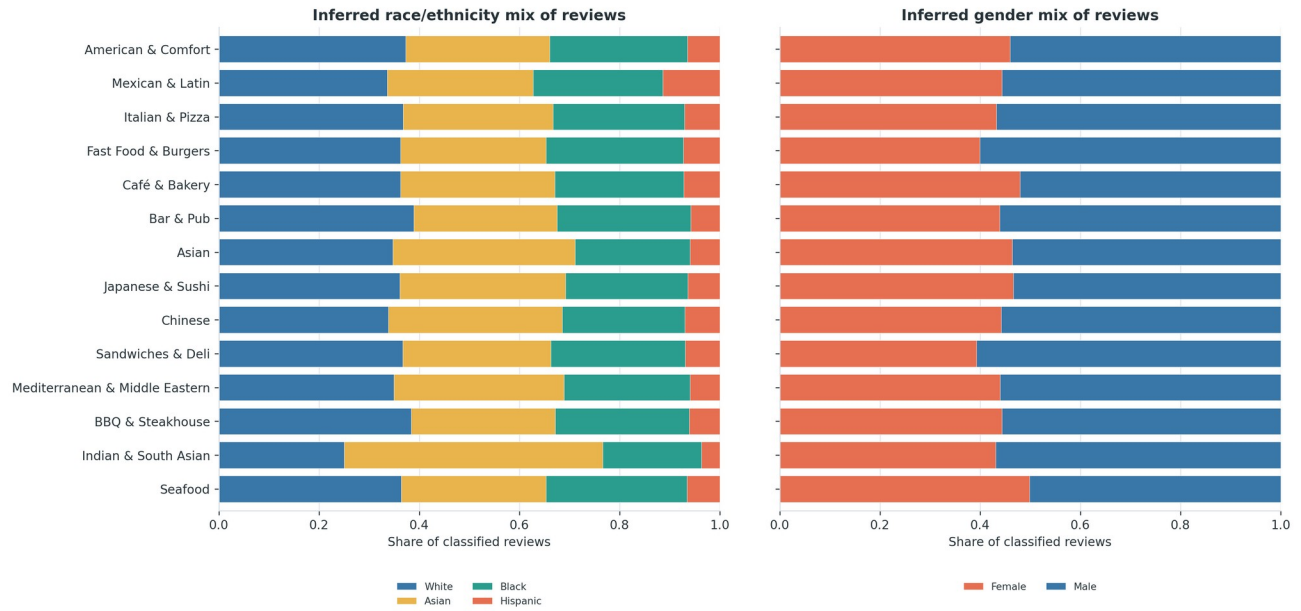


Figure 9. Demographic composition within the classified subsets. Shares are conditional on a classified name-inference output and are not estimates of customer populations.

Modeling population and chronological split

The recommendation experiment deduplicates user–restaurant pairs and retains reviewers with at least four distinct restaurants. For each eligible user, the newest interaction is test, the second-newest is validation, and all older interactions form training.

Split	Interactions
Training	772,170
Validation	156,261
Test	156,261

This chronological boundary also governs behavioral aggregates, LLM corpora, graph edges, model selection, and GraphRAG evidence. The next article examines how semantic features are constructed without allowing validation or test review text to cross that boundary.